

Instructions:

- You may use any calculator except a TI-nspire.
- Put away all phones, laptops, and course materials (other than your formula sheet).
- Show all of your work, and use correct notation. Full credit will not be given without sufficient supporting work.
 - If you use a theorem, rule, definition, etc., tell me that you're using it!
- Round all numeric answers to at least 2 decimal places.
- I have attached 2 blank pages at the end of the exam. Feel free to tear these off and use them as scratch paper.
- You are awesome, and I believe in you!

1. (6 pts.) Suppose you flip 3 fair coins and observe whether each is heads or tails. List the outcomes in the sample space, S .

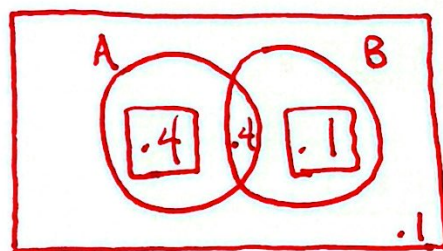
$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

2. (18 pts.) Suppose that at a certain college, 80% of freshmen are taking English 101 this semester, 50% are taking Mathematics 101, and 40% are taking both of these courses. Define $A = \{\text{the freshman is taking English 101}\}$ and $B = \{\text{the freshman is taking Mathematics 101}\}$.

- (a) (6 pts.) What is the probability that a randomly selected freshman is taking at least one of these two courses?

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \quad [\text{Additive Law}] \\ &= 0.80 + 0.50 - 0.40 \\ &= 0.90 \end{aligned}$$

- (b) (6 pts.) What is the probability that a randomly selected freshman is taking exactly one of these two courses (Hint: drawing a Venn diagram might help)?



$$\begin{aligned} &P(\text{exactly one of these courses}) \\ &= P(A \setminus B \cup B \setminus A) \\ &= P(A \setminus B) + P(B \setminus A) \quad [\text{Additive Law for disjoint events}] \\ &= 0.4 + 0.1 \\ &= 0.5 \end{aligned}$$

- (c) (6 pts.) Are the events A and B independent? Justify your answer mathematically.

$$P(A \cap B) = 0.40 = (0.80)(0.50) = P(A)P(B).$$

Hence, A and B are independent.

3. (6 pts.) A group of four undergraduate and eight graduate students are available to fill certain student government posts. If five students are to be randomly selected from this group, find the probability that exactly three graduate students will be among the five chosen. Simplify your answer!

$$P(\text{exactly 3 graduate students}) = \frac{\binom{8}{3} \binom{4}{2}}{\binom{12}{5}} \approx .4242$$

4. (16 pts.) A department store sells only two types of phones: iPhones and Androids. Suppose that 75% of the phones sold are iPhones, and 25% of the phones sold are Androids. According to the manager, 3% of iPhones are returned, while 8% of Androids are returned. Consider a randomly selected phone purchase.

(a) (8 pts.) What is the probability that the phone is returned? Let $R = \{\text{phone is returned}\}$.

Given: $P(\text{iPhone}) = 0.75$

$P(\text{Android}) = 0.25$

$P(R|\text{iPhone}) = 0.03$

~~$P(R|\text{iPhone})$~~ $P(R|\text{Android}) = 0.08$

Want: $P(R)$

$$\begin{aligned} P(R) &= P(R|\text{iPhone}) \cdot P(\text{iPhone}) + P(R|\text{Android}) \cdot P(\text{Android}) \quad [\text{LTP}] \\ &= (0.03)(0.75) + (0.08)(0.25) \\ &= 0.0425 \end{aligned}$$

(b) (8 pts.) Given that the phone is returned, what is the probability that the phone is an Android?

Equivalently,
using
Bayes'
Rule!

$$\begin{aligned} P(\text{Android} | R) &= \frac{P(\text{Android} \cap R)}{P(R)} \quad [\text{def. of conditional probability}] \\ &= \frac{P(R|\text{Android}) \cdot P(\text{Android})}{P(R)} \quad [\text{Multiplicative Law}] \\ &= \frac{(0.08)(0.25)}{0.0425} \\ &\approx 0.4706 \end{aligned}$$

5. (8 pts.) For $Y \sim \text{Geom}(p)$, verify that $E(Y) = \frac{1}{p}$ using the moment-generating function of Y .

$$\begin{aligned} E(Y) &= \left. \frac{d}{dt} m(t) \right|_{t=0} = \left. \left[\frac{d}{dt} \frac{pe^t}{1-(1-p)e^t} \right] \right|_{t=0} \\ &= \left. \frac{pe^t[1-(1-p)e^t] - [- (1-p)e^t pe^t]}{[1-(1-p)e^t]^2} \right|_{t=0} \quad [\text{Quotient Rule}] \\ &= \left. \frac{pe^t}{[1-(1-p)e^t]^2} \right|_{t=0} \\ &= \frac{p}{p^2} \\ &= 1/p \end{aligned}$$

6. (20 pts.) A printing business has two telephone lines. The probability distribution for Y , the number of lines in use at a specified time, is as follows:

y	0	1	2
$p(y)$	.25	.45	.30

- (a) (6 pts.) Calculate the mean of Y .

$$\begin{aligned} E(Y) &= \sum_{y=0}^2 y p(y) = 0(.25) + 1(.45) + 2(.30) \\ &= 0 + .45 + .60 \\ &= 1.05 \end{aligned}$$

- (b) (6 pts.) Calculate the variance of Y .

$$\begin{aligned} V(Y) &= \sum_{y=0}^2 (y - 1.05)^2 p(y) = (0 - 1.05)^2 (.25) + (1 - 1.05)^2 (.45) + (2 - 1.05)^2 (.30) \\ &= .275625 + .001125 + .27075 \\ &= .5475 \end{aligned}$$

- (c) (4 pts.) Calculate $E(2Y - 4)$.

$$E(2Y - 4) = 2E(Y) - 4 = 2(1.05) - 4 = -1.9$$

- (d) (4 pts.) Calculate $V(2Y - 4)$.

$$V(2Y - 4) = (2)^2 V(Y) = 4(.5475) = 2.19$$

7. (8 pts.) Dr. Hopper spontaneously decides to give up the math-lete life to pursue a basketball career (HA). Suppose he takes 15 3-point shots and has a 15% chance of making each one. Assume each shot is independent. What is the probability that he makes more than 4 shots? Before calculating the probability, be sure to define your random variable, along with its distribution.

Let X = # of 3-point shots that Dr. Hopper makes.

Then $X \sim \text{Bin}(15, .15)$.

$$P(X > 4) = 1 - P(X \leq 4) \quad [\text{Complement Rule}]$$

$$= 1 - \text{binomcdf}(15, .15, 4)$$

$$\approx .0617$$

↑
out ∴

(18 pts.) Choose two of problems 8-10 to complete. If you work on more than two problems, indicate which ones you want graded; otherwise, your instructor may select which ones to grade.

8. (9 pts.) Prove the following result: $V(Y) = E(Y^2) - [E(Y)]^2$.

$$\begin{aligned}
 V(Y) &= E[(Y - \mu)^2] \quad (\text{by def. of variance}) \\
 &= E[Y^2 - 2\mu Y + \mu^2] \\
 &= E(Y^2) - 2\mu E(Y) + \mu^2 \quad (\text{using properties of expected values}) \\
 &= E(Y^2) - 2[E(Y)]^2 + [E(Y)]^2 \quad (\mu = E(Y)) \\
 &= E(Y^2) - [E(Y)]^2. \quad \square \quad (\text{combining like terms})
 \end{aligned}$$

9. (9 pts.) Suppose A and B are mutually exclusive events and $P(B) > 0$. Prove that

$$P(A|A \cup B) = \frac{P(A)}{P(A) + P(B)}.$$

$$\begin{aligned}
 P(A|A \cup B) &= \frac{P(A \cap (A \cup B))}{P(A \cup B)} \quad (\text{def. of conditional probability}) \\
 &= \frac{P(A)}{P(A \cup B)} \quad (A \cap (A \cup B) = A) \\
 &= \frac{P(A)}{P(A) + P(B)} \quad \square \quad (\text{Additive Law for disjoint events})
 \end{aligned}$$

10. (9 pts.) Show that the moment-generating function of $Y \sim \text{Poisson}(\lambda)$ is $e^{\lambda(e^t-1)}$.

$$\begin{aligned} m(t) &= E(e^{ty}) = \sum_{y=0}^{\infty} e^{ty} \cdot \frac{\lambda^y}{y!} e^{-\lambda} \\ &= \sum_{y=0}^{\infty} \frac{(\lambda e^t)^y}{y!} e^{-\lambda} \\ &= e^{-\lambda} \underbrace{\sum_{y=0}^{\infty} \frac{(\lambda e^t)^y}{y!}}_{\substack{\text{power series} \\ \text{expansion of} \\ e^{\lambda e^t}}} \\ &= e^{-\lambda} e^{\lambda e^t} \\ &= e^{\lambda(e^t-1)}. \quad \square \end{aligned}$$